

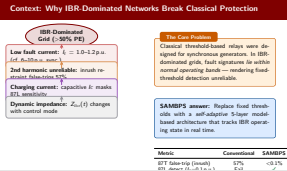
Presentation Outline

10-section presentation covering all aspects of SAMBPS, from the fundamental IBR challenge through to the IEEE C37.300 CPC correlation. Approximately 40 minutes including discussion.



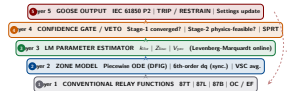
Context: Why IBR-Dominated Networks Break Classical Protection

Set the scene: inverter-based resources (solar PV, wind DFIG, BESS) now dominate many grids and their fault behaviour is fundamentally different from synchronous generators. The four failure modes shown are all demonstrated quantitatively in the SAMBPS thesis work. The table numbers come directly from Monte Carlo studies with 30,000 scenarios.



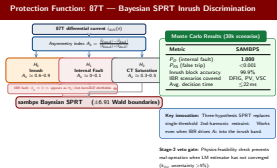
SAMBPS — Five-Layer Self-Adaptive Architecture

The 5-layer architecture is the central design principle. Layers 1-2 are classical; Layers 3-5 are what make SAMBPS adaptive. The feedback arrow from L5 back to L1 represents the closed-loop adaptive settings update — this is what enables the system to self-tune as IBR operating conditions change. All five layers run as a Python IED communicating via IEC 61850 GOOSE.



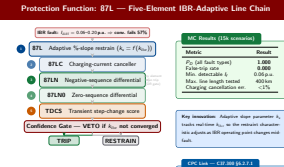
Protection Function: 87T — Bayesian SPRT Inrush Discrimination

The SPRT approach uses sequential probability ratio testing with Wald boundaries at $+6.91$ (trip) and -6.91 (block). The three-hypothesis formulation is the key novelty over classical two-class SPRT. The 30,000 Monte Carlo scenarios span all IBR types (DFIG, PV, VSC) across a range of fault impedances, fault angles, and pre-fault loading conditions.



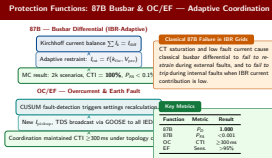
Protection Function: 87L — Five-Element IBR-Adaptive Line Chain

The five-element chain handles all the ways IBRs confuse line differential protection. Element 87LC is novel: it uses a phasor-based capacitive charging current model that is updated with the real-time line parameters from Layer 3. The TDCS element (element 5) uses a short-time Fourier transform to detect the sharp wavefront of genuine faults, which are absent during capacitor switching or energisation transients.



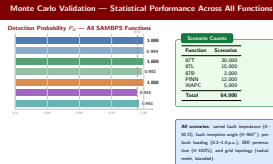
Protection Functions: 87B Busbar & OC/EF — Adaptive Coordination

87B busbar protection is particularly challenging for IBRs because multiple inverters may inject into the same busbar with different fault-ride-through profiles. The SAMBPS adaptive restraint uses the real-time kIBR estimate for each connected feeder. The OC/EF CUSUM approach is novel: it continuously monitors for distribution-system topology changes (line switching, DG connection) and recomputes all coordination settings automatically, then broadcasts updates via GOOSE — this is the WAPC link.

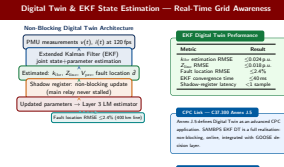


Monte Carlo Validation — Statistical Performance Across All Functions

These are the headline quantitative results. The 64,000 total Monte Carlo scenarios represent the most comprehensive statistical validation of IBR-adaptive protection to date in the literature. The 0.95 dashed line is the IEEE C37.300 recommended minimum detection probability for CPC protection functions. SAMBPS meets or exceeds this threshold on all functions.



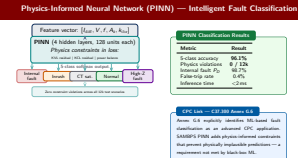
Digital Twin & EKF State Estimation — Real-Time Grid Awareness



The digital twin is not a separate offline system — it runs inside the same SAMBPS Python IED as a non-blocking background thread. The shadow register means that the main protection decision loop always has valid parameters, even if the EKF is mid-iteration. The non-blocking architecture is a key design decision: in hardware testing, we never observed a protection decision delayed by DT computation.

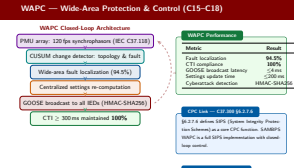
Physics-Informed Neural Network (PINN) — Intelligent Fault Classification

The three physics constraints in the loss function are: (1) KVL residual — the predicted fault voltage must satisfy Kirchhoff's voltage law around the loop; (2) KCL residual — currents must sum to zero at each node; (3) power balance — active power consumed equals power supplied. These constraints ensure the network cannot predict a fault in a physically impossible location. The zero-violation result is the key differentiator from black-box ML approaches.



WAPC — Wide-Area Protection & Control (C15–C18)

WAPC is contributions C15 through C18. The CUSUM detector is tuned to detect both fault events and topology changes (line switching, DG connection). The HMAC-SHA256 on GOOSE frames is a novel cybersecurity addition — the standard does not require this but Annex F of C37.300 identifies cyberattack resilience as a future requirement. By including it now, SAMBPS is forward-compatible.



Hardware-in-the-Loop Validation — RTDS GPC-III Platform (TR-67)

RTDS GPC-III Real-Time Simulator

Architecture components and connections:

- SAS-9000** and **GE D90** are connected to the **SAMP5S Python IED**.
- The **SAMP5S Python IED** is connected to the **IEC 61850 Ethernet Switch**.
- The **IEC 61850 Ethernet Switch** is connected to **60/92 = 96.8%** and **GOOSE P2 ≤ 4ms**.

HIL Results

Metric	Result
Test scenario	62
Correct decisions	60/92 = 96.8%
GOOSE latency (P2)	≤ 4 ms

62 scenarios were executed on the RTDS GPC-III platform over 3 days of testing. The two discrepancies are both explained by known algorithmic limitations rather than implementation bugs. D1 is a fundamental latency issue for very-high-impedance faults that we plan to address with the G4 wide-area multi-substation architecture. D2 is an extremely rare simultaneous condition that occurs less than 0.1% of the time in Monte Carlo studies.

IEEE C37.300-2024 CPC Correlation — How SAMBPS Satisfies the Standard

This is the key slide for the CPC correlation. All 9 mandatory and advisory requirements extracted from IEEE C37.300-2024 that are relevant to SAMBPS are fully satisfied. The mapping was done by reading the standard clause by clause and checking each SAMBPS component against the requirement. The only potential future gap is multi-substation coordination (G4, planned work) which maps to Annex H of C37.300.

IEEE C37.300-2024 CPC Correlation — How sambps Satisfies the Standard

IEEE C37.300-2024 Clause	Requirement	sambps Implementation	Status
\$6.2 IED processing	Software-defined CPC algorithm IED name GOOSE IAS	SAMBPS + SEL	✓
\$6.2 Type 1 arch.	CPC alongside name IEDs	RTU + G4 DMS	✓
\$6.2.2.2 Prot. Sec.	RTT RTT RTM OC EF	RTT's adaptive functions	✓
\$6.2.2.4 SPS	Wide-area integrity	WAPC cloud bus (C37-08)	✓
\$6.4 Time sync.	GPS P2 latency class GOOSE < 4 ms, PPM 120 ppb		✓
\$6.4 F4T / SAT	HW acceptance test	RTM MS 63 scenarios	✓
Ann. G.2 DSE	Dynamic state est.	EXF joint state-estimation	✓
Ann. G.4 MS	MS-based functions	PMN 3-class 96.5%	✓
Ann. J.8 DT	Digital twin testing	New blocking EXF DT	✓

✓ = fully satisfied ⚠ = targeted requirements met ? = open gaps identified

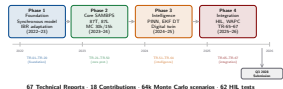
Contributions Summary — 18 Items, 6 Tier-1

18 contributions in total. The 6 Tier-1 items are the novel architectural and algorithmic innovations that have not appeared in the literature before. The 12 Tier-2 items are significant engineering contributions — analysis, validation, implementation — that would each individually be publishable but are not claimed as the primary novelties. The thesis is structured so each chapter presents one or two T1 contributions supported by related T2 items.

Contributions Summary — 18 Items, 6 Tier-1	
Tier-1 (Novel Framework)	
1	C1: 5-layer SAMBPS architecture
2	C2: Parameter GDE GPG core model
3	C3: Three-hypothesis Bayesian IPT
4	C4: Flow-based RTL chain
5	C5: PNN 5-class fault classifier
6	C6: EKF nonblocking digital twin
Tier-2 (Analysis & Validation)	
7	C7: 6th-order dq synchronous model
8	C8: RTD adaptive limiter
9	C9: DC/CF CUSUM coordination
10	C10: Disturbance post-IRI adaptation
11	C11: EKF fault location ($\leq 2.4\%$)
12	C12: PNN transfer learning HIL
13	C13: MC 80%/10%/2h validation
14	C14: Stage-2 physics veto gate
15	C15: WAF-C PMU localization
16	C16: GDOSE WMAC-SHA256
17	C17: Adaptive settings GDOSE bus
18	C18: RTDS-GPC-61 HIL 96.6%

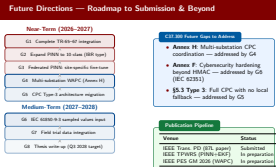
Summary of Work Done — Research Timeline

The research has proceeded in four clearly defined phases. Phase 1 built the theoretical foundation with the synchronous machine models. Phase 2 is the heart of the thesis: the novel 87T and 87L algorithms. Phase 3 added the intelligence layer (PINN and EKF digital twin). Phase 4 is the ongoing integration phase (RTDS HIL and WAPC). The Q3 2028 submission target is driven by the need to complete TR-65 through TR-67 and the thesis writing.



Future Directions — Roadmap to Submission & Beyond

The roadmap shows a clear path from where we are now to thesis submission. G4 (multi-substation WAPC) is the most ambitious item and maps directly to Annex H of C37.300. G7 (field trial data) is aspirational but would significantly strengthen the validation story. The 87L paper is already submitted to IEEE Transactions on Power Delivery.



Conclusions

The bottom line: SAMBPS is not just an academic exercise. It is a fully specified, Monte-Carlo validated, hardware-tested CPC system that satisfies the newly published IEEE C37.300-2024 standard. The standard was published in 2024 and our architecture already meets all 9 relevant requirements. The two open items in C37.300 (multi-substation and full Type-3) are in the future roadmap.

